Prabhavit Gupta

LLM Application Engineer: Multimodal Pipelines, Structured Extraction & Evals

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit | github.com/prab-gupta | Venice, Italy (Remote / US Eastern Overlap Ready)

Summary

LLM Application Engineer with 3+ years of production experience architecting end-to-end AI pipelines, multimodal extraction workflows, and evaluation harnesses. Deep hands-on expertise building resilient structured extraction systems that transform unstructured inputs (PDFs, images, emails, complex HTML) into validated schema-conforming database records. Skilled in leveraging Google Gemini, Anthropic Claude, and OpenAI models with rigorous token budgeting, prompt versioning, confidence scoring, and automated golden-set regression testing. Proven track record operating in fast-paced product studios and agency environments, collaborating seamlessly with cross-functional teams across US hours to ship observable, cost-effective serverless pipelines.

Experience

FlowSpace.co • *Lead Developer & AI Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Architected and operated end-to-end multimodal data ingestion pipelines, extracting structured entities and booking parameters from mixed formats (PDFs, invoices, images, spreadsheets) for 4,500+ active enterprise users.
- Implemented strict JSON Schema validation and fallback mechanisms on model outputs (Google Gemini, Anthropic Claude), eliminating extraction errors and preventing malformed records from persisting to databases.
- Designed and maintained offline evaluation harnesses using golden benchmark datasets to test precision, recall, and edge-case handling before deploying model or prompt updates.
- Profiled latency and token expenditure across model tiers, introducing semantic caching and optimized prompt framing to slash inference expenses while maintaining high extraction fidelity.
- Built asynchronous event-driven queues with retry mechanisms, idempotency keys, and dead-letter alerting to handle long-running extraction jobs reliably.
- Automated continuous integration and test verification using GitHub Actions and Docker, ensuring zero regressions on live production endpoints.
- *Tools: Python, Google Gemini API, Anthropic Claude API, OpenAI API, TypeScript, PostgreSQL, Supabase, Docker, Linux*

Urlink • *Project Manager & Solutions Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Delivered custom client onboarding automation and structured data extraction workflows for European client accounts.
- Established automated API validation pipelines connecting third-party platforms to PostgreSQL databases, guaranteeing data consistency.
- Managed project delivery end-to-end: gathered client requirements, designed technical architectures, and delivered complete operational runbooks.
- *Tools: Python, TypeScript, Claude Code, REST APIs, Supabase*

Independent Technical Consultant • *AI Automation & Integration Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Developed and deployed custom automated integration test scripts, webhook listeners, and web scraping pipelines using Python, TypeScript, and n8n.
- Replaced brittle manual verification processes with automated regression test flows, featuring robust retry mechanisms, error alerting, and structured reporting.
- Documented testing procedures, test coverage matrices, and API contracts for client engineering teams.
- *Tools: Python, TypeScript, n8n, Supabase, Webhooks, REST APIs*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Integrated LLM API workflows and email verification services, debugged responsive web applications, and built test scenarios for core user funnels.
- *Tools: OpenAI API, Sendgrid, Figma, REST APIs*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft Copilot Studio Agent Academy • Microsoft: Introduction to building AI agents • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** European Long Term Residence Permit (Italy), unrestricted EU work authorization. Available immediately.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (beginner).
- **Availability:** Available for full-time / retainer commitments; comfortably overlaps US Eastern Time.

Core Skills

- **LLM Engineering & Multimodal AI:** Google Gemini (multimodal document/image extraction), Anthropic Claude, OpenAI, structured JSON outputs, prompt engineering, context management, taxonomy classification.
- **Evaluation & Quality Systems:** Offline eval harnesses, golden dataset benchmarking, regression test suites, token budget optimization, hallucination prevention, confidence thresholds.
- **Backend & Cloud Systems:** Python, TypeScript, Node.js, Cloud Functions, serverless architectures, RESTful API design, PostgreSQL, Supabase, Redis, Docker containerization.
- **Developer Workflows & Reliability:** Claude Code, GitHub Actions CI/CD, asynchronous job queues, idempotent retries, comprehensive technical runbooks.